

SMART INDIA HACKATHON 2026



SMART INDIA
HACKATHON
2026

- **Problem Statement ID:** SIH26028
- **Problem Statement Title :**
Dynamic Forecast of Expected
Time of Arrival (ETA) for Coaching
Trains
- **Theme :** Disaster Management
- **PS Category :** Software
- **Team ID :** 104
- **Team Name :** Blitz



Blitz

DYNAMIC ETA FORECAST FOR COACHING TRAINS

 SMART INDIA HACKATHON 2026



LIVE TRAIN DATA

Position
Speed
Current Delay
Next Station

➔



HISTORICAL + ROUTE DATA

Delay Patterns
Timetable
Route
Distance

➔



ML PREDICTION

Feature Engineering + XGBoost

➔



DYNAMIC ETA

Upcoming Stations
Continuously Updated
Arrival Prediction

➔



LIVE VISUALIZATION

Route Map
Station-wise ETA
Delay Insights

<u>Existing Live-Status Systems</u>	<u>TrainETA</u>
Show current train status	Predict future arrival
Current delay information	Live + historical intelligence
Schedule / expected timings	Station-wise ML forecasts
Track where the train is	Track + predict where it will be
Status updates	Continuously updated ETA

TrainETA

HomeTrack TrainSearchSaved TrainsNotifications

Live Indian Railways Tracking & Smart ETA

Track smarter. Arrive better.

Search trains, monitor live running status, and get reliable arrival estimates across Indian Railways.

Find & Track Your Train

Enter train number, name, or choose origin and destination stations.

Train number / name

From

To

Date

e.g. 12951 or Rajdhani

Origin station

Destination station

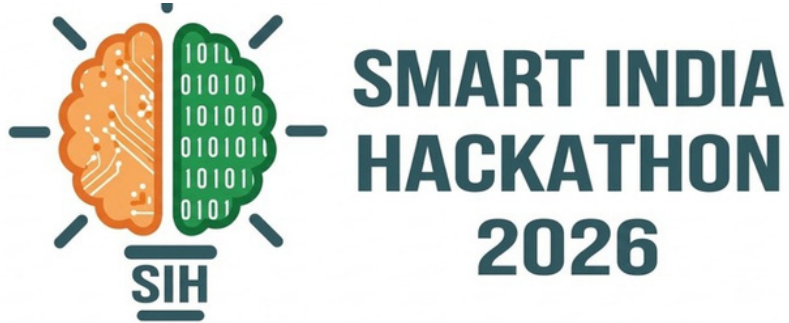
19-09-2026

Search by train number, or choose a route pair to get live status and estimated arrival.

Track Train

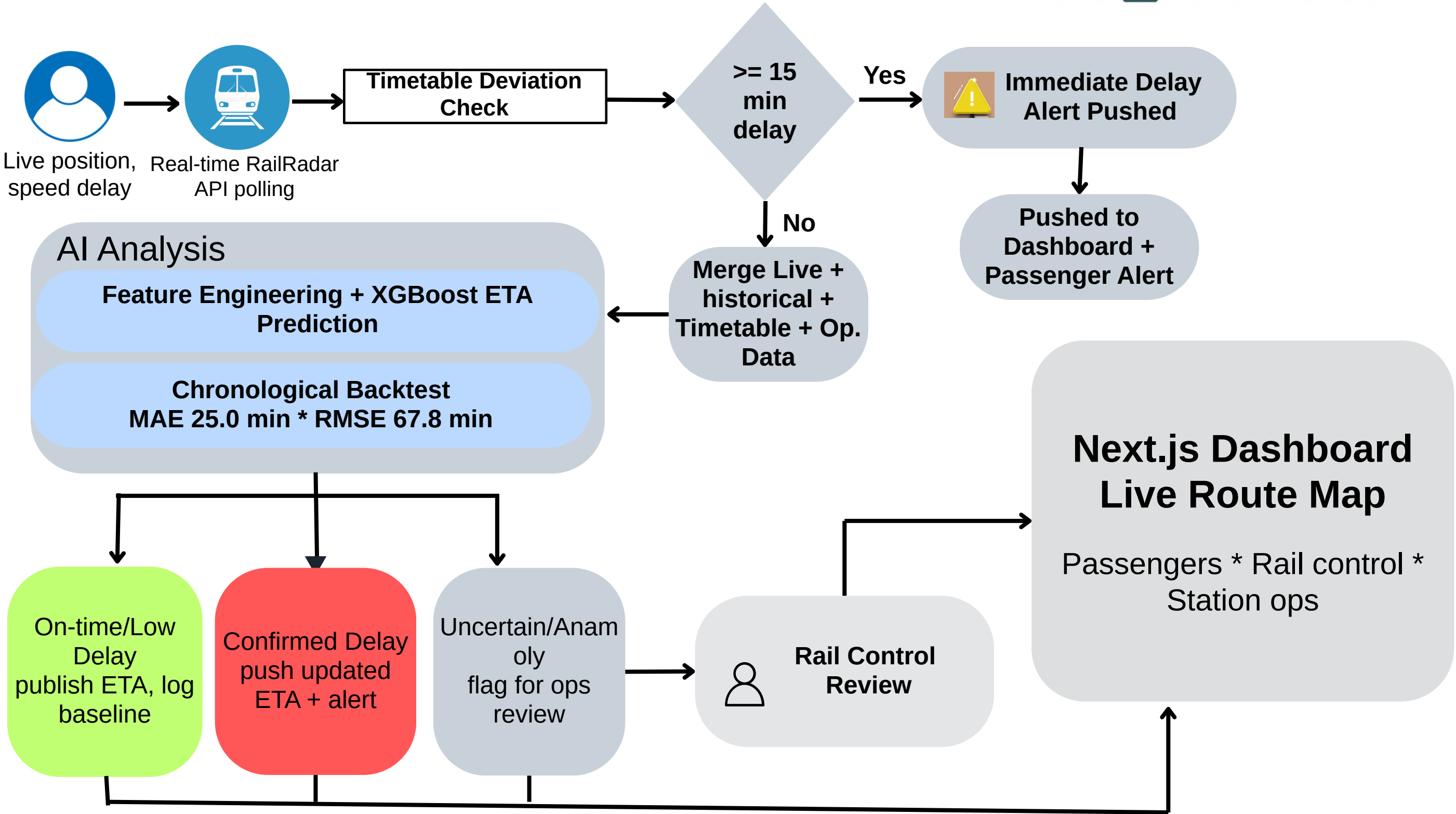


TECHNICAL APPROACH

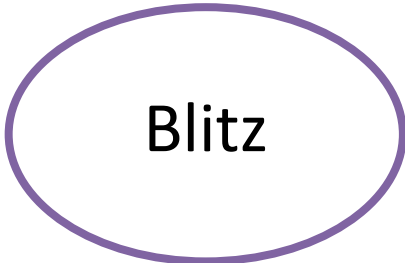


Techstack Used:

- 1.Next.js + React + TypeScript
- 2.Tailwind CSS
- 3.Python + FastAPI
- 4.XGBoost
- 5.PostgreSQL
- 6.RailRadar API
- 7.MapLibre GL+OpenStreetMap
- 8.Git + GitHub + VS Code



Model Performance (evaluation set)			
57,465	25.0 min	67.8 min	+8.9%
journeys evaluated	Mean Absolute Error	RMSE	vs. current-delay-only



FEASIBILITY AND VIABILITY



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TECHNICALLY FEASIBLE

- ✓ Technical
FastAPI + PostgreSQL + XGBoost + MapLibre support a modular architecture.
- ✓ Data
Historical railway data, timetable/route information and live train-data integration support the prototype
- ✓ Deployment
Separate frontend/backend architecture enables cloud deployment
- ✓ Scalable
Modular data sources allow additional routes, trains and stations to be incorporated.

CHALLENGE

- Live API availability
- Historical vs live mismatch
- Delay propagation
- Changing operations
- Intermediate stations
- ML leakage



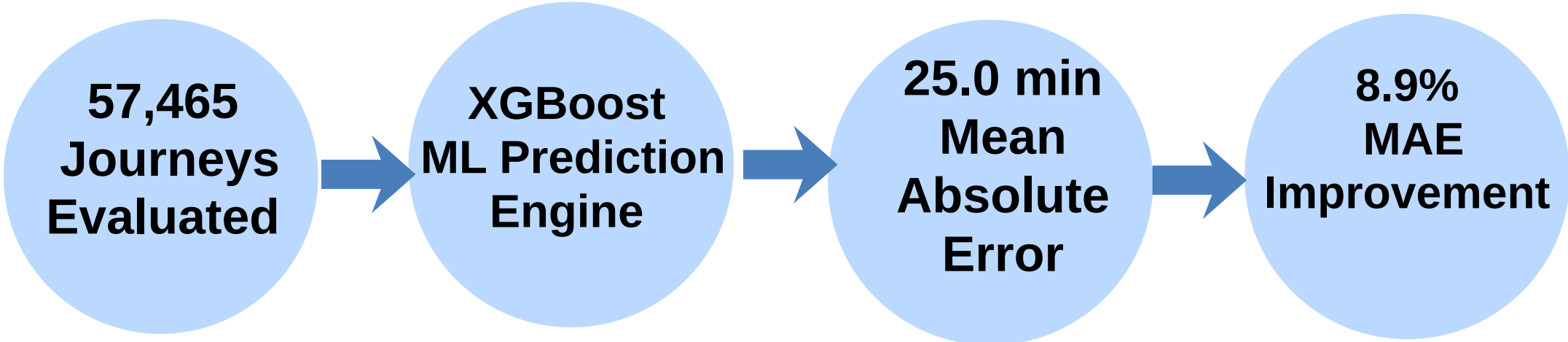
SOLUTION

- Modular live-data integration
- Route & station alignment
- Historical + ML prediction
- Dynamic operational adjustment
- Distance-based fallback
- Chronological evaluation

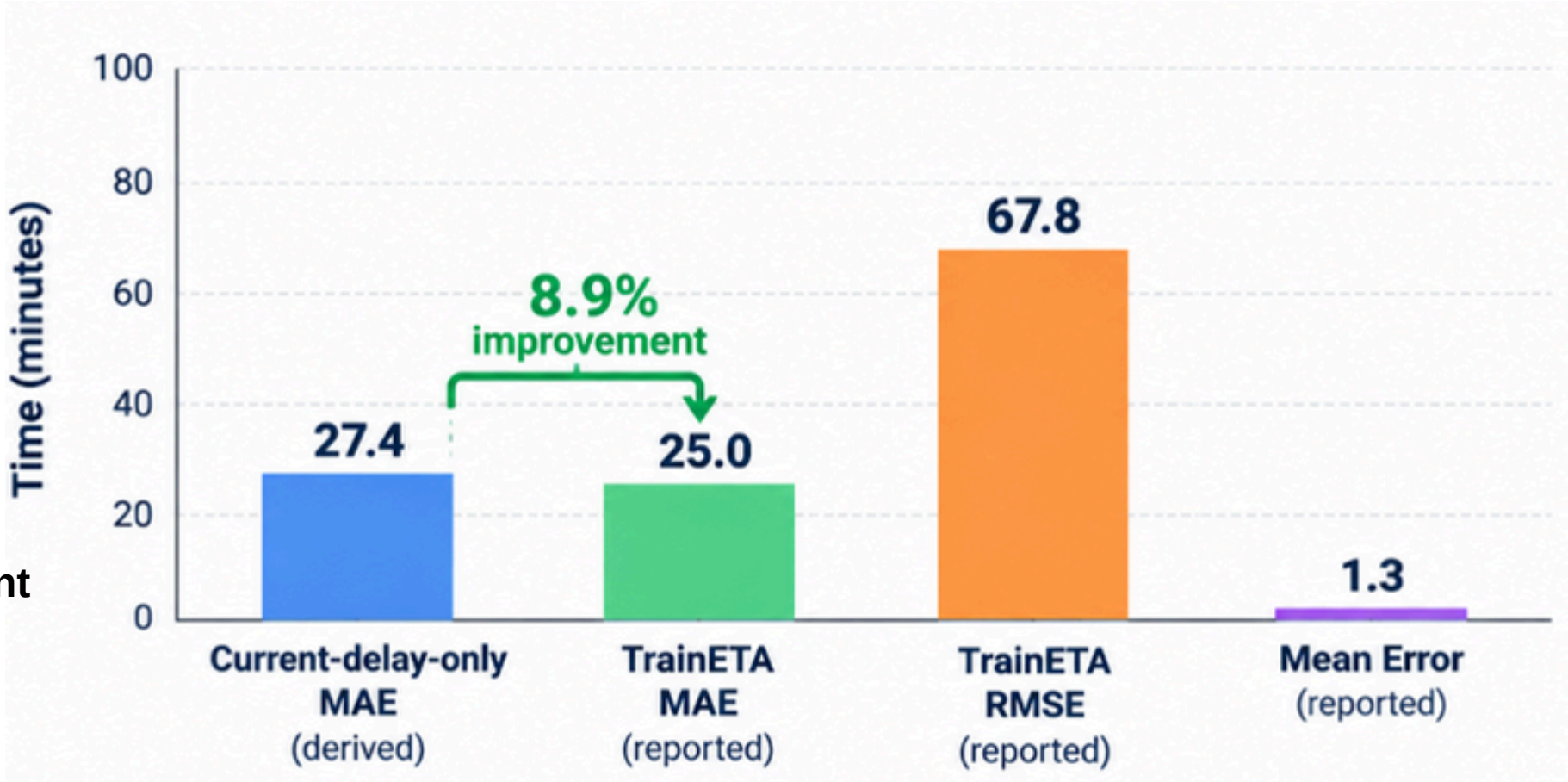
SCALABILITY

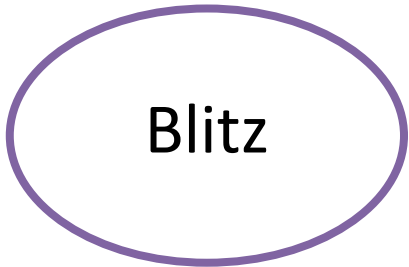


PROTOTYPE VALIDATION

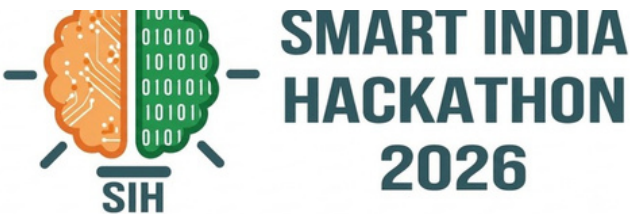


Chronological evaluation prevents future-data leakage.





IMPACT AND BENEFITS



WHO BENEFITS?

Passengers

Realistic arrival information
Better journey planning
Reduced uncertainty

Railways

Better arrival visibility
Improved station planning
Better passenger information

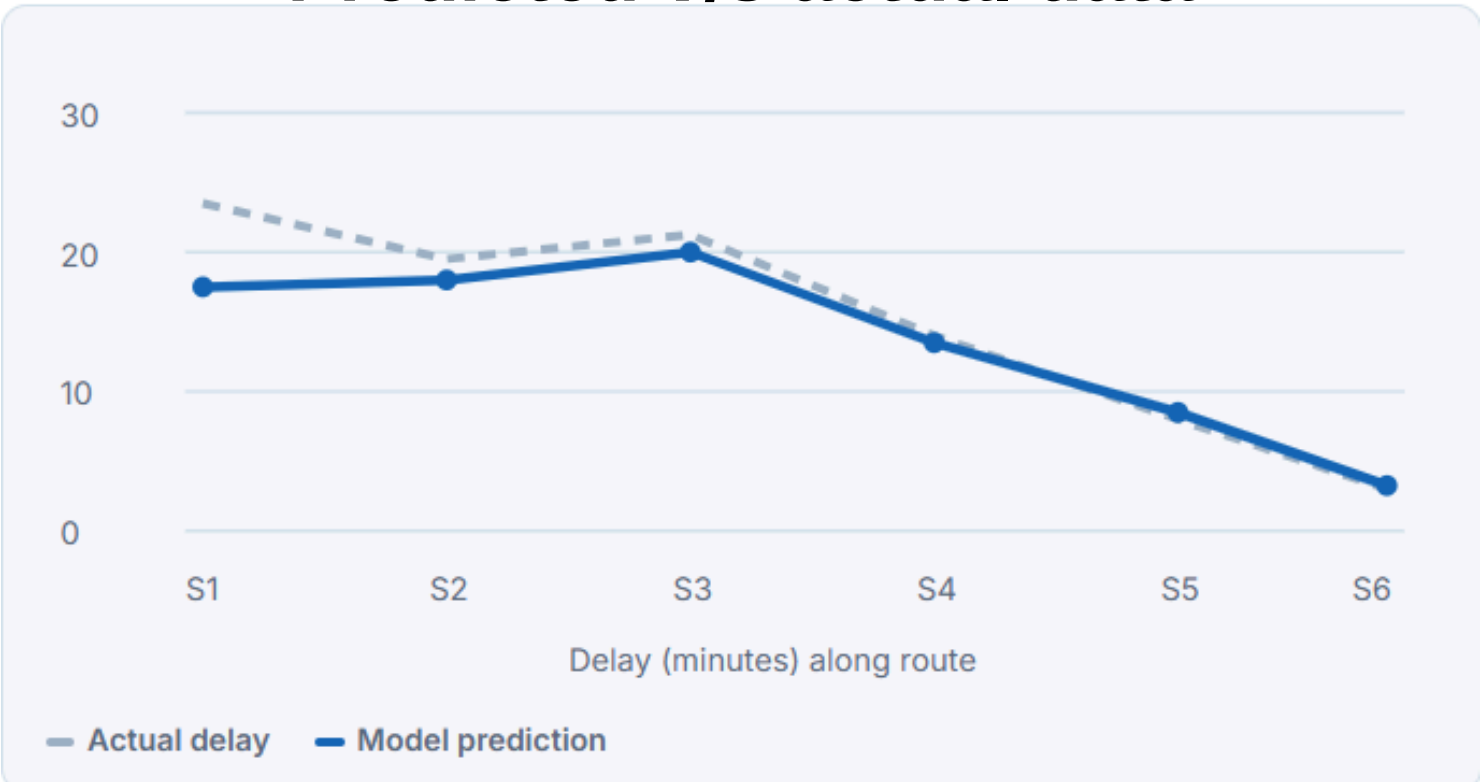
Stations

Dynamic train visibility
Delay propagation insights
Decision-support information

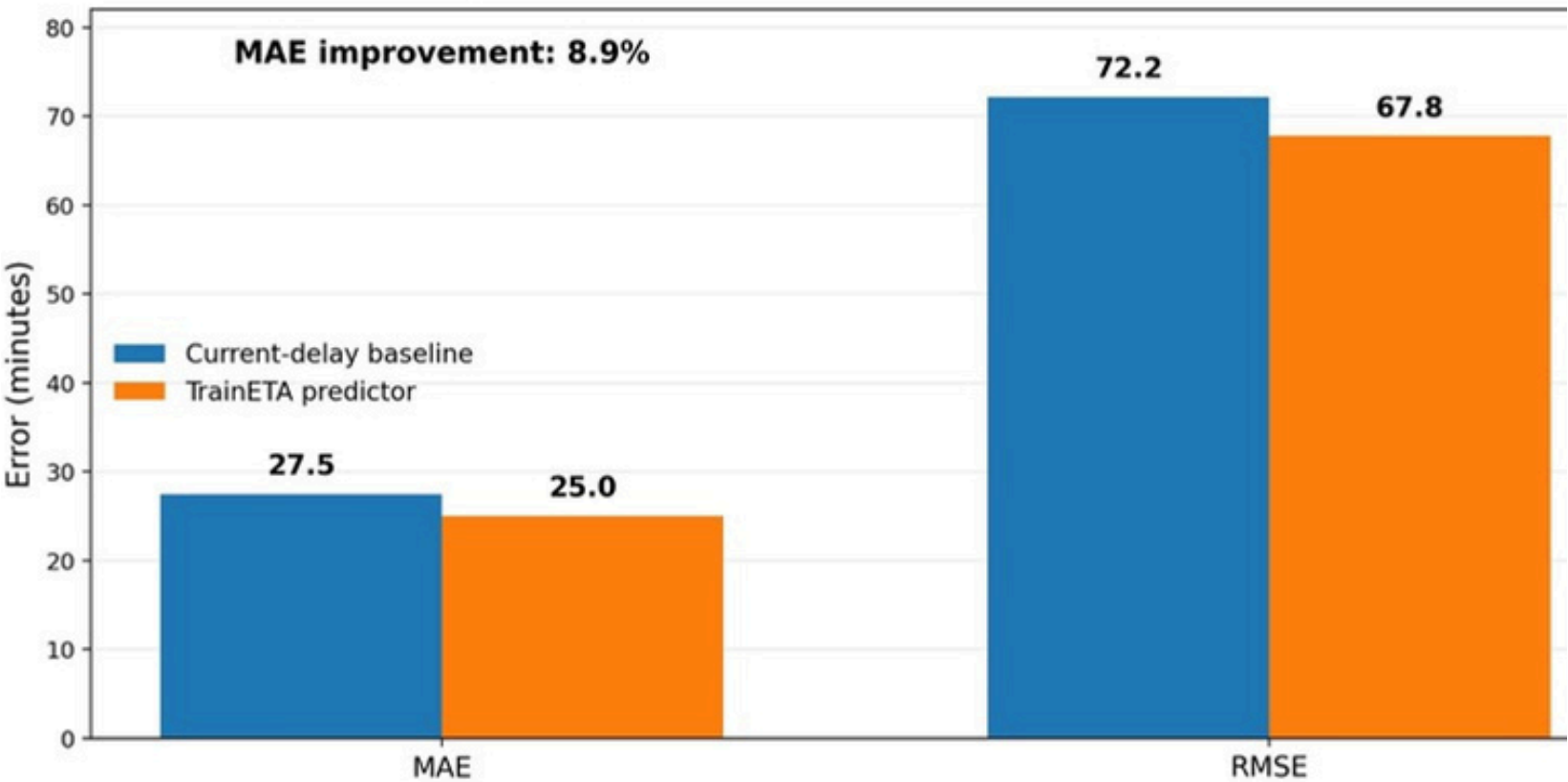
Operations

Operational monitoring
Resource planning
Reduced arrival uncertainty

Predicted v/s actual data



TrainETA Prediction Error Comparison



What are the impacts?



SOCIAL IMPACT

1. Better passenger information
2. Reduced uncertainty during delays
3. Improved access to real-time train status



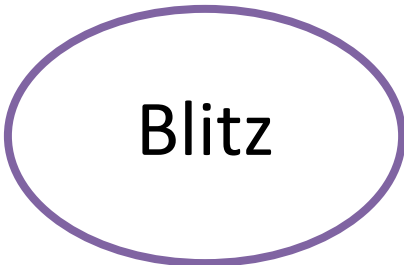
OPERATIONAL IMPACT

1. Dynamic visibility of train movement
2. Delay propagation insights
3. Data-driven decision support

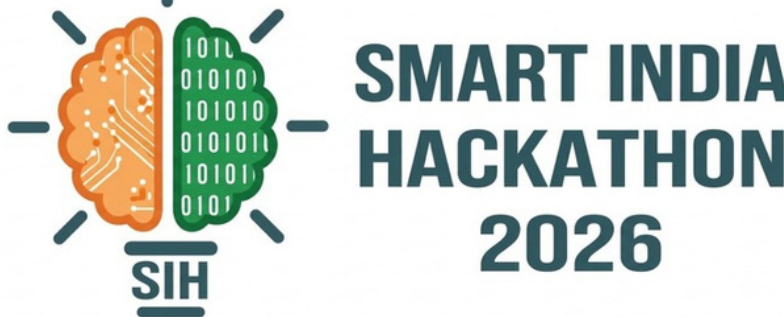


ECONOMIC IMPACT

1. Better utilization of passenger & station resources
2. Reduced uncertainty around delayed arrivals
3. Potentially more efficient railway operations



RESEARCH AND REFERENCES



Research / Data Sources

- Smart India Hackathon 2026 — Problem Statement SIH26028: Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains.
- Indian Railway historical route & delay dataset — used for historical delay analysis and model development.
- RailRadar — live train-data source used in the working prototype.
- XGBoost — machine-learning framework used for ETA prediction.
- FastAPI Documentation — backend REST API implementation.
- PostgreSQL Documentation — railway data storage and retrieval.
- MapLibre / OpenStreetMap / OpenFreeMap — railway route visualization.

Experimental Evidence

Evaluation journeys	57,465
MAE	25.0 minutes
RMSE	67.8 minutes
Historical component vs current-delay-only	8.9% MAE improvement

References

- 1) RailRadar Live Train Status API Documentation — supports your live position, delay, speed and next-halt integration
- 2) XGBoost research paper — “A Scalable Tree Boosting System” — the research foundation for your XGBoost model.
- 3) MapLibre GL JS Documentation — supports your interactive railway map implementation.